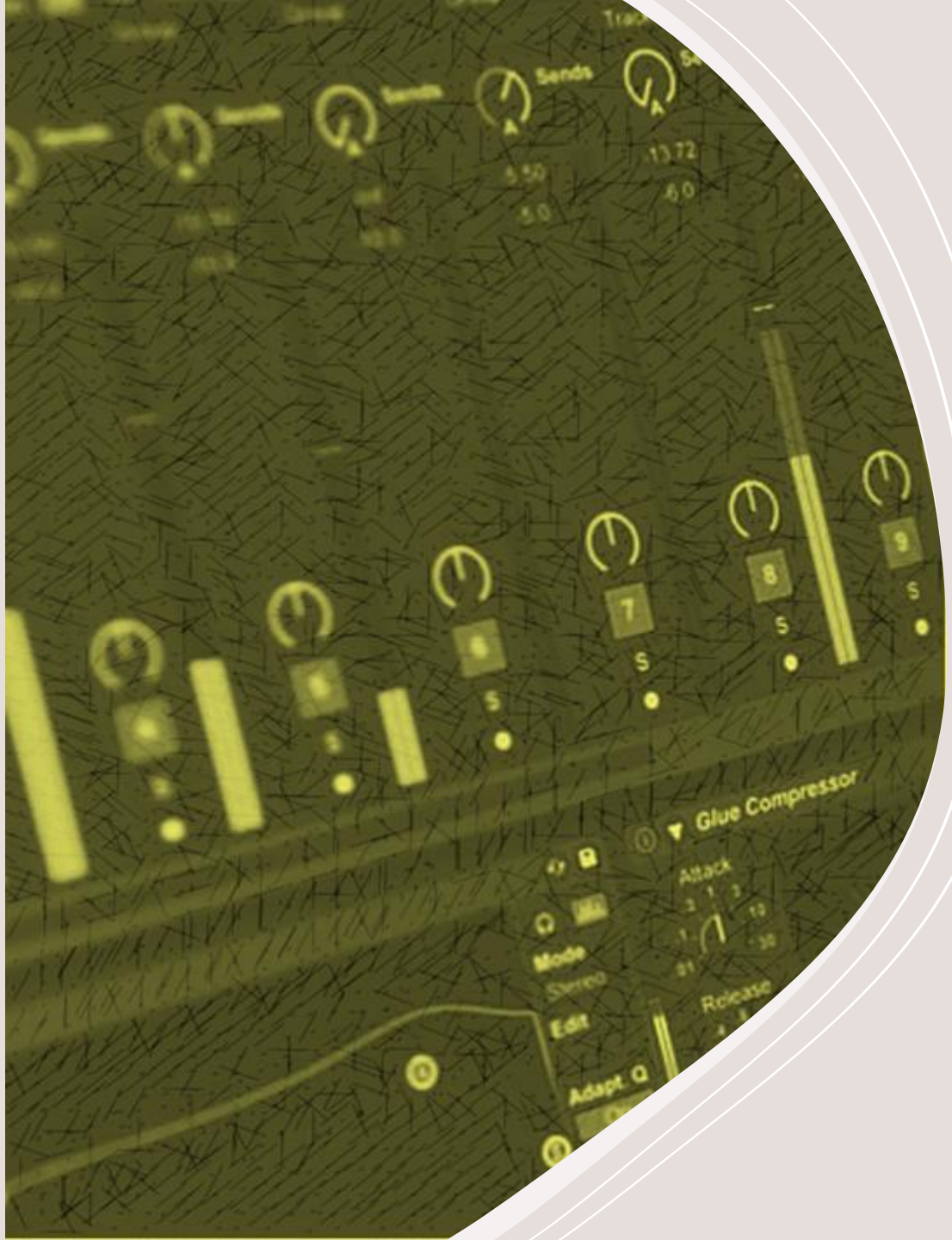


香港中文大學
The Chinese University of Hong Kong

Music in the Digital World

AIST2010 Introduction to Computer Music

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Department of Computer Science and Engineering



Sound and Music in Real Life

Digitizing sound and Music Data

Musical Instruments

Music Hardware and Software

Outline

A bit of physics, music theory, and audio engineering in this lecture!

Sound and Music in Real Life

Sound and Music in Real Life

Vibration and sound

- The ***vibration*** of air particles from the source into your ears
  - Or via other media (e.g., water)
  - Vibrations detected by human auditory system
    ➔ "audio" signals

Different vibrations give rise to different kinds of sounds

- Some are *harmonious*, some are *noise*, some are high, some are low

VIBRATION

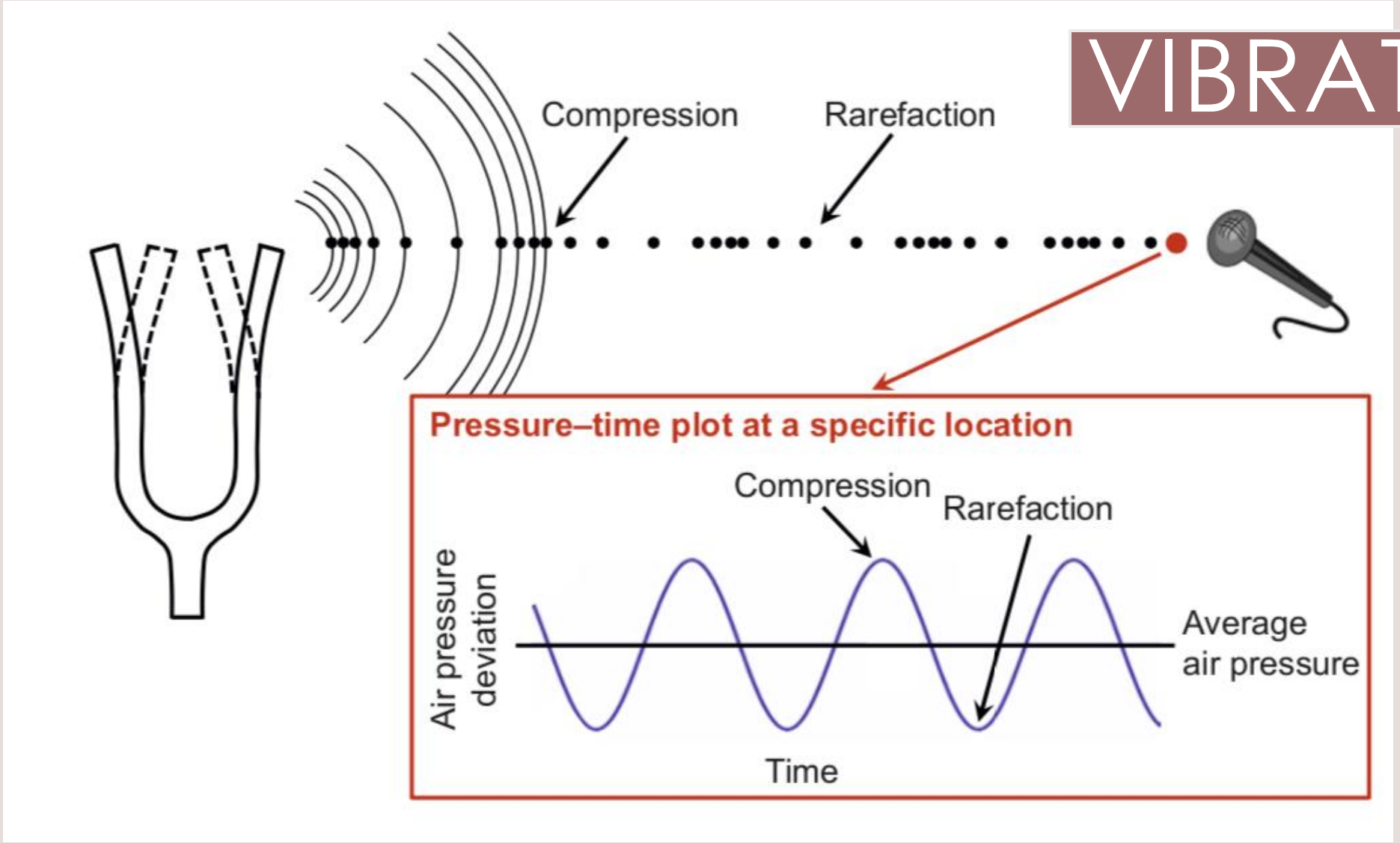


Image from: ***Fund. of Music Processing**, p.20*

For periodic signals, frequency measured in ***Hz***
→ cycle per second (cps)

Video from: https://www.youtube.com/watch?v=YV6XAfZcCCE (1:04:53)



Sound and Music in Real Life

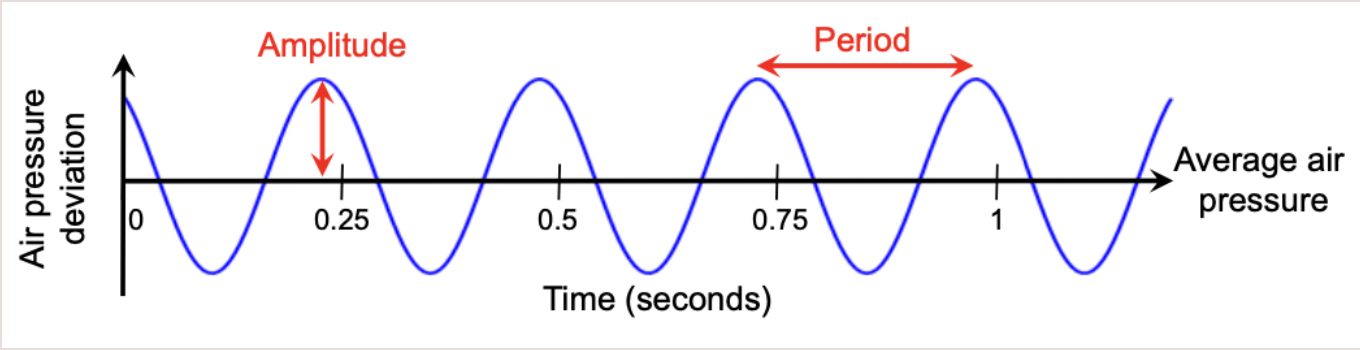
Music

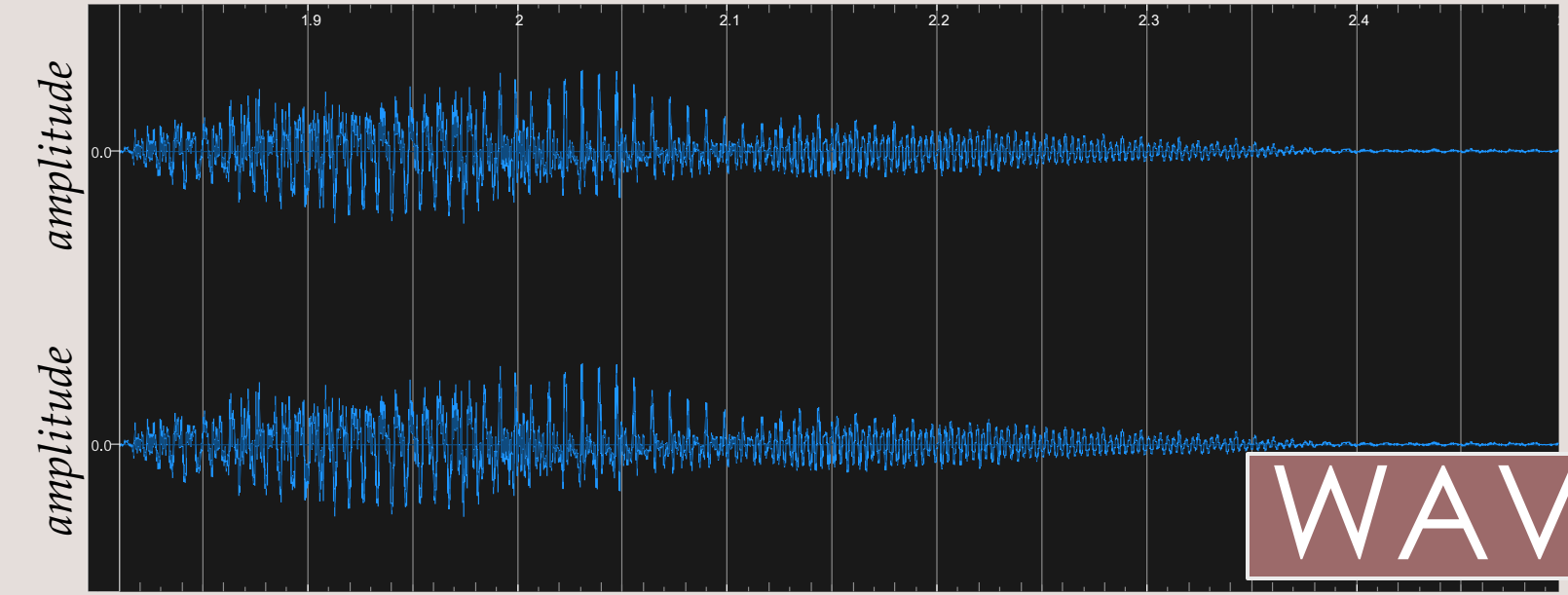
- A *clever combination* of sounds and voids

Visualizing sounds in computers

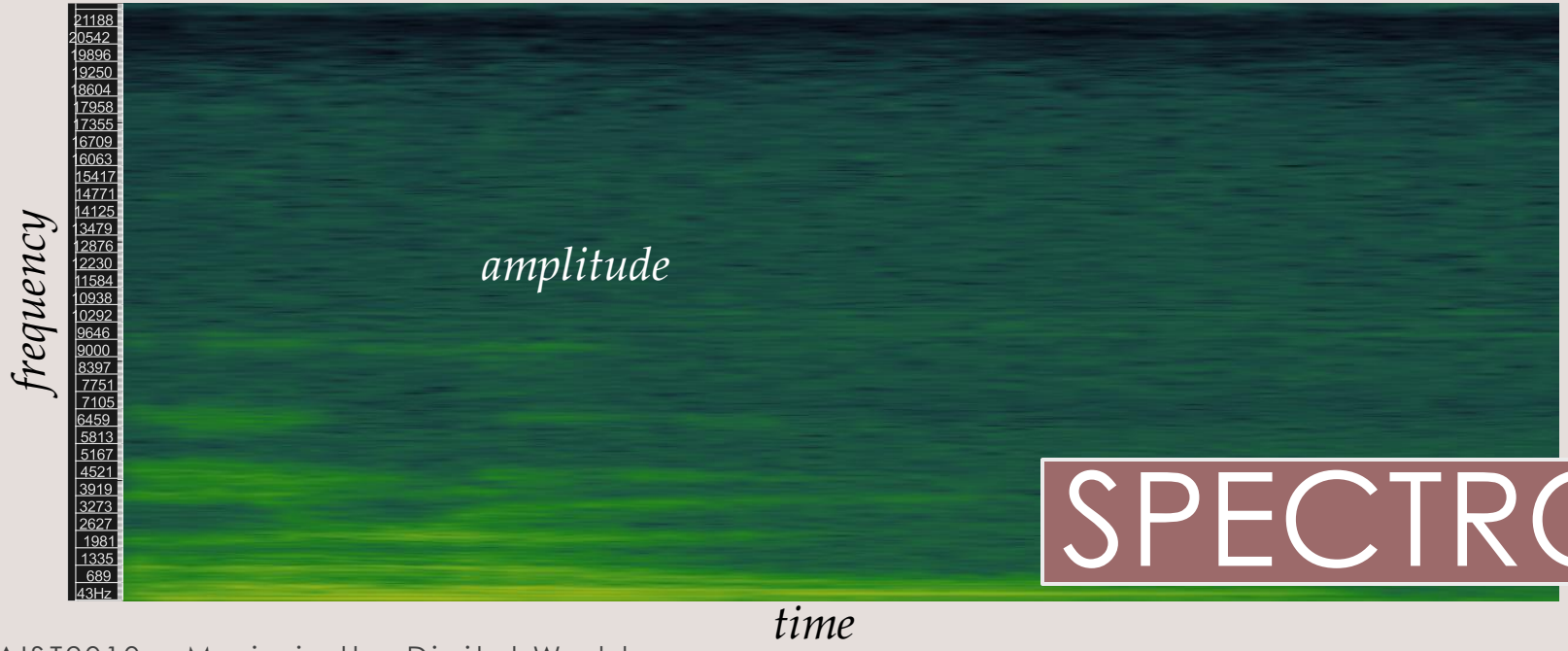
- It's always an ***abstraction***, e.g.,
  - Waveform (*amplitude vs. time*)
  - Spectrum (*frequency vs. amplitude. vs. time*)

Image from: ***Fund. of Music Processing***, *p.21*





WAVEFORM



Note: There are other ways to represent the spectrum too

SPECTROGRAM

Basic Periodic Waveforms

A pitch is heard for ***periodic*** waveforms (at certain frequencies)

- Sine wave
- Cosine wave
  - Phase-shifted sine wave
- Triangular wave
- Sawtooth wave
- Square wave

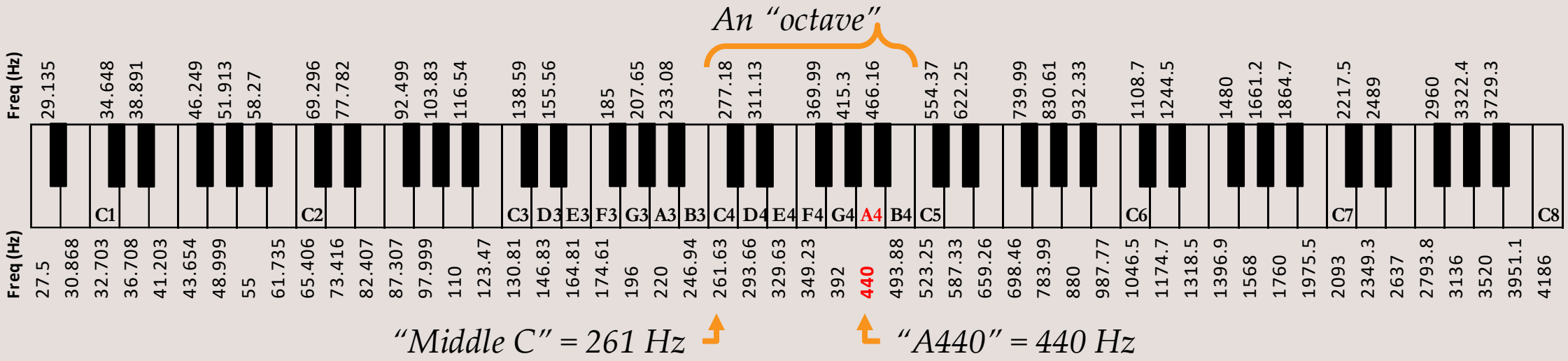


Image from: https://moogfoundation.org/learning-synthesis/synthesis-fundamentals/

Commonly used in vintage synthesizers (i.e., analog electronics)

Measuring Pitch

Pitch (***frequency***) range on a piano keyboard



- Double frequency = "octave" higher
  - The next occurrence of the same ***letter-name*** in music

Measuring Pitch

Pitch is perceived in a ***logarithmic*** manner

- Feeling of doubling frequency
  - Change from 110 to 220 Hz is similar to that from 440 to 880 Hz ➔ "an octave higher"
  - # of octaves $= \log_2(f_2/f_1)$

Perceptual scales of pitch

- Mel-scale
- Critical bands (Bark scale)
- Equivalent rectangular bandwidth (ERB) rate
- log Frequency

} *Empirical scales!*

Human Ear and Psychoacoustics

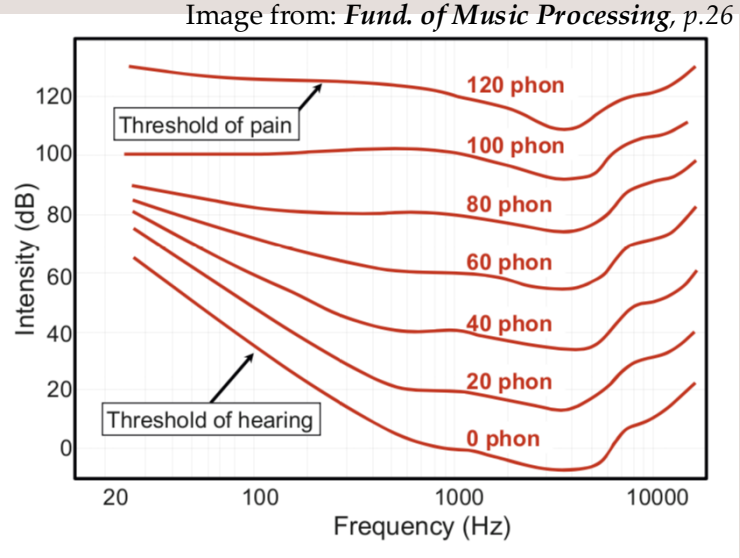
Human sound perception is ***subjective***

- General range: ~20 Hz – ~20,000 Hz (20 kHz)
- Hearing range narrows with age, especially at high frequencies
  - How well do animals hear? *https://lsu.edu/deafness/HearingRange.html*

Human is more sensitive to ~2 to 4 kHz

- A not-so-flat "*frequency response*"
- For the sake of human communication?
- Perception of loudness is also affected by age

Image from: ***Fund. of Music Processing***, *p.26*



Equal loudness contours

Measuring Loudness

Source	Intensity	Intensity level	× TOH
Threshold of hearing (TOH)	10^{-12}	0 dB	1
Whisper	10^{-10}	20 dB	10^{2}
Pianissimo	10^{-8}	40 dB	10^{4}
Normal conversation	10^{-6}	60 dB	10^{6}
Fortissimo	10^{-2}	100 dB	10^{10}
Threshold of pain	10	130 dB	10^{13}
Jet take-off	10^{2}	140 dB	10^{14}
Instant perforation of eardrum	10^{4}	160 dB	10^{16}

Image from: ***Fund. of Music Processing***, *p.25*

Loudness perception is also ***logarithmic***

Sound power = (amplitude)^2

- Sound intensity = sound power per unit area (*watt per sq. meter*)

"Threshold of hearing" (TOH) vs "Threshold of pain" (TOP)

- $I_{TOH} \stackrel{\text{def}}{=} 10^{-12}\ \text{W/m}^2$
- TOP is roughly 10^{13} times of the intensity of TOH

- *Note: Power depends on distance!*

Measuring Loudness

Source	Intensity	Intensity level	× TOH
Threshold of hearing (TOH)	10^{-12}	0 dB	1
Whisper	10^{-10}	20 dB	10^{2}
Pianissimo	10^{-8}	40 dB	10^{4}
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Instant perforation of eardrum	10^{4}	160 dB	10^{16}

Image from: ***Fund. of Music Processing***, *p.25*

Decibel (dB) is a logarithmic unit to ***measure the ratio***

$$\mathrm{dB}(I) \stackrel{\text{def}}{=} 10 \cdot \log_{10}\left(\frac{I}{I_{TOH}}\right)$$

- i.e., TOP is 130 dB when TOH is 0 dB

- This value relative to TOH is also called the *sound pressure level* (SPL), i.e., $\mathrm{dB}_{\mathrm{SPL}}$

Measuring Loudness

The ***amplitude*** (which we discuss more in computer audio) can also be represented in dB

- Since it is only a ratio, an assumption is needed for the comparison ground

$$A_{dB}(x) = 20 \cdot \log_{10} x$$

*Note: This term is **20** because intensity is amplitude **squared***

Do not mess up with…

- dB_A or dB(A) represents loudness perceived by human ear
  - Compensation using the A-weighting basing on equal-loudness curves

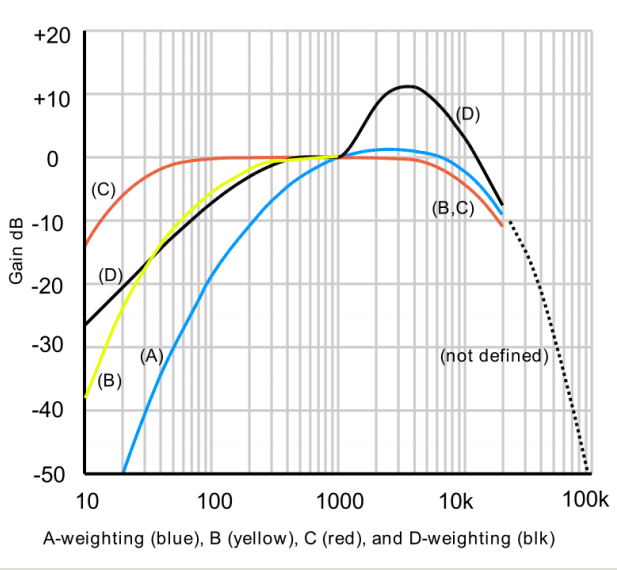


Image from: https://en.wikipedia.org/wiki/A-weighting

Measuring Loudness

Decibel is only a ratio!

- Occasionally, the decibel amplitude has 0 dB the loudest, and –ve values for lower amplitude

- From the formula…
  - $A_{dB}(x) = 20 \cdot \log_{10} x$
  - 6 dB = double the amplitude



Aspects of Music

When sounds and rests are *cleverly* put together, we have music with…

- *Melody*
  - combination of pitches (at different time)
- *Rhythm*
  - combination of long and short sounds
- *Harmony* (chords)
  - combination of pitches (at the same time)
- *Harmonic progression*
  - combination of chords (at different time)
- *Dynamics*
  - the expression in loudness
- *Structure*
  - how music sections are arranged
- *Texture*
  - the amount of different/similar sounds at times
- *Timbre*
  - different shapes of waveforms

  ➔ different spectrum

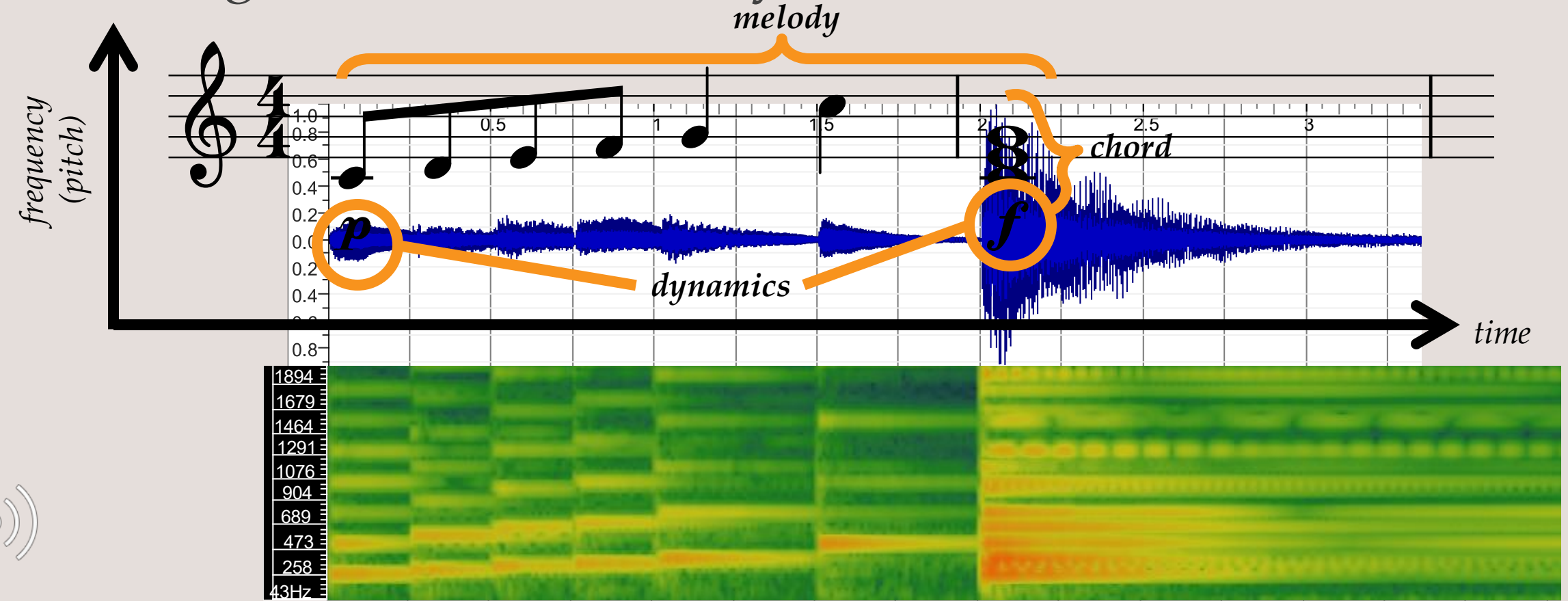
There can be more!

It's a musical collage I made from YouTube recordings…



Aspects of Music

Reading music scores is easy!



Digitizing Sound and Music Data

Representing Sounds

Audio representation

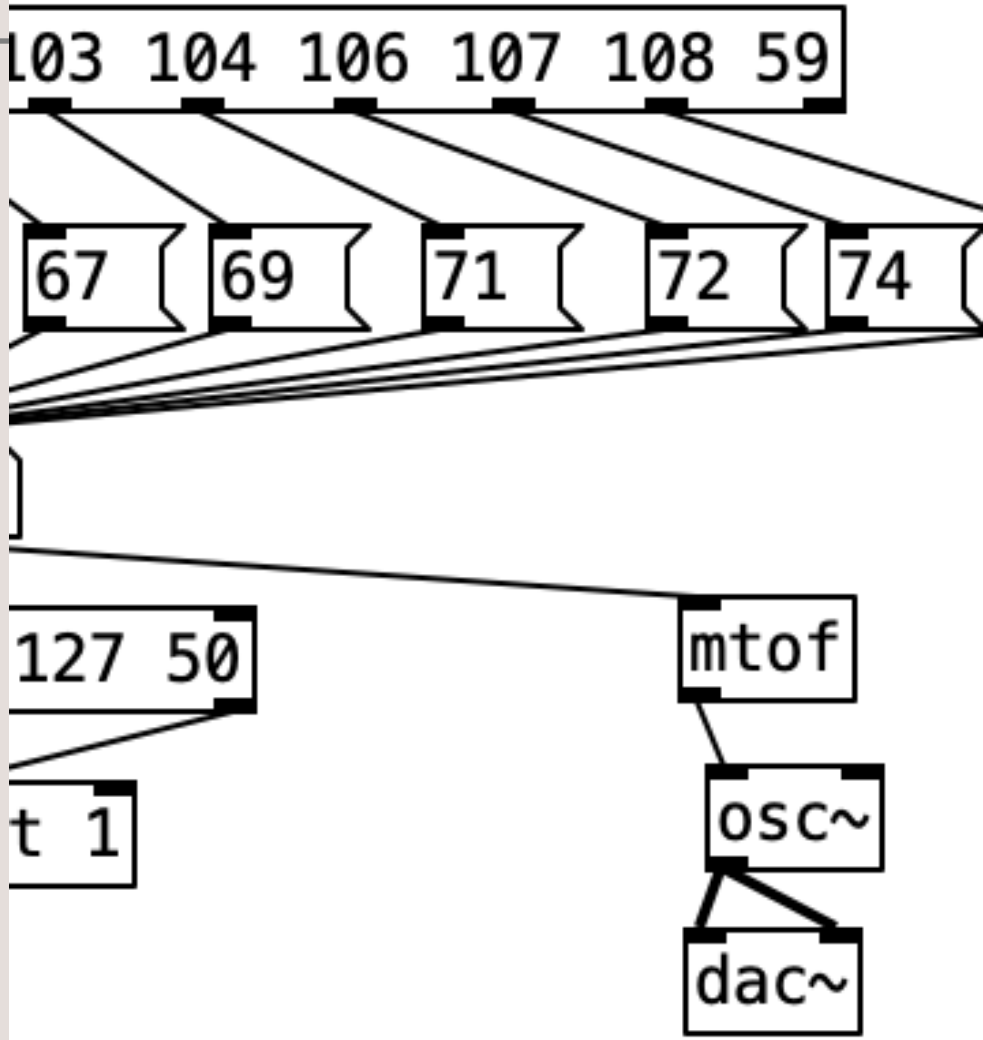
- The waveform shape
- Faithful representation of *what can be heard*
  - Recording and playback
- To improve audio representation, research is on the enhancing sound fidelity



Representing Sounds

Symbolic representation

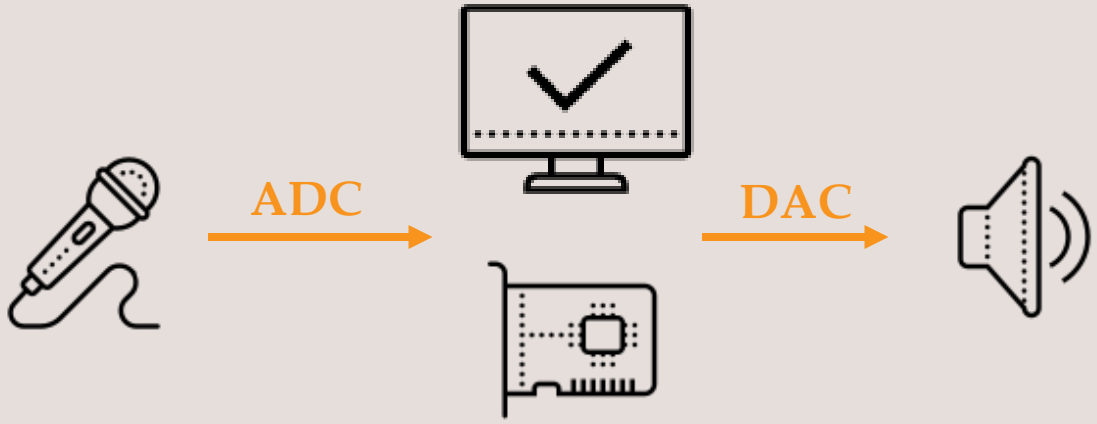
- The musical idea
- Reconstruction of *music data* (e.g., melody, rhythm, etc.)
  - Analysis and synthesis
- To improve symbolic representation, research is on the analysis and synthesis methods



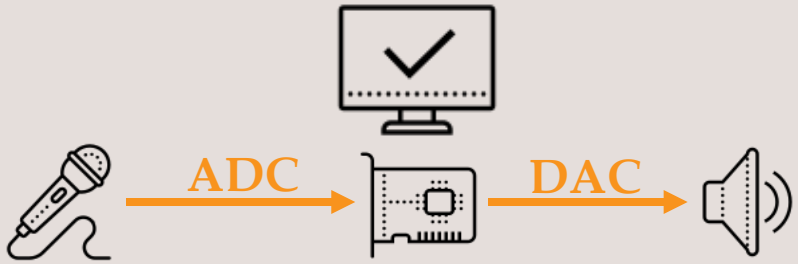
Storing Audio Data: "Audio Representation"

To represent audio signals in the *digital world*, ***digitization*** is needed

- ***ADC*** (analog-digital conversion) for recording
- ***DAC*** (digital-analog conversion) for playing back



Storing Audio Data: "Audio Representation"



Sampling: the representation in time

- The higher sampling rate the better
  - CD quality: 44.1 kHz
    ➔ 44,100 samples every second

Quantization: the representation in amplitude

- The more dynamic levels the better
  - CD quality: 16-bit
    ➔ 2^{16} = 65,536 different levels of amplitude

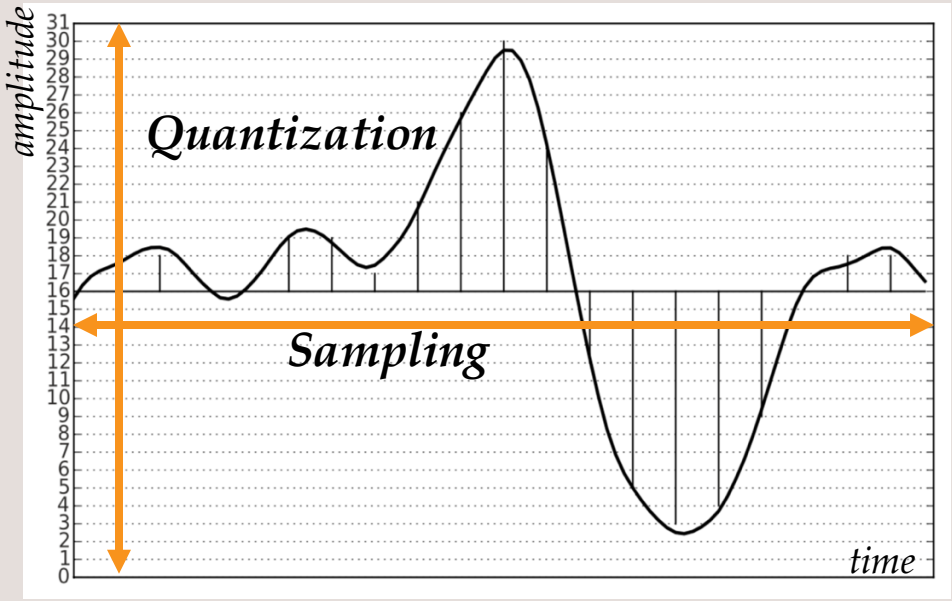


Image from: ***Comp. Music Instruments***, *p.21*

Sampling: Sampling Rate

How many samples per second is *the best*?

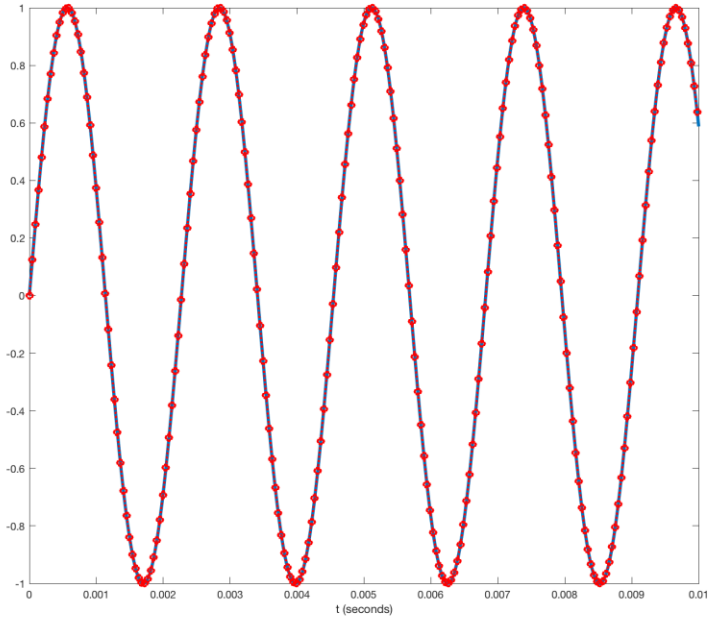
- Tradeoff between storage requirements vs. sound quality

Sampling Theorem

- Signal reconstructable if frequency is not higher than…
  - ***Nyquist frequency***: $\Omega = F_s/2$, where F_S is the sampling rate
  - Otherwise, artifacts called aliasing will form → *an extra frequency component will appear!*

Common rates

Analog telephone: 8,000 Hz	Pro digital video equipment: 48,000 Hz
Low quality audio: 22,050 Hz	DVD-Audio, Blu-ray audio: 96,000 Hz
Audio CD, MP3: **44,100 Hz**	Super-Audio CD: 2,822,400 Hz



440Hz signal at 22,050Hz sampling rate

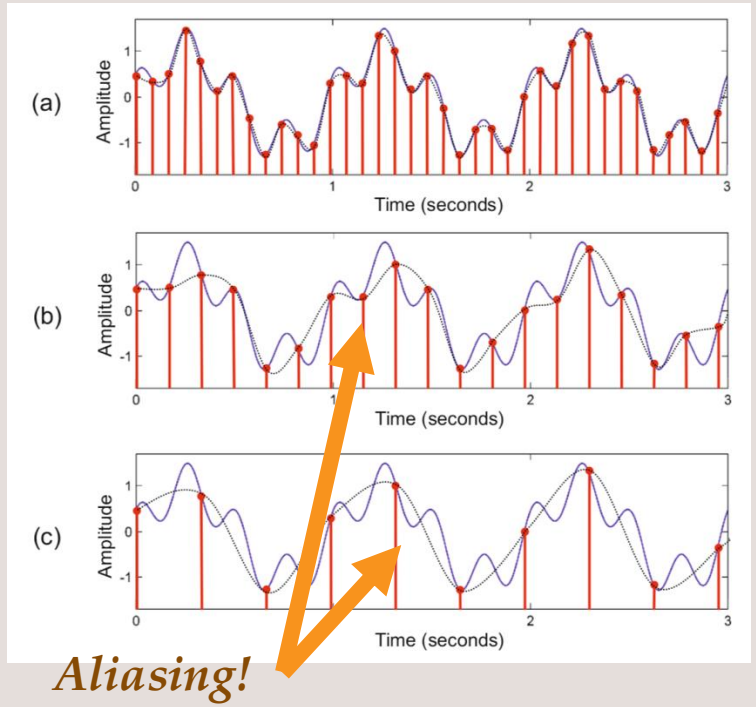


Image from: ***Fund. of Music Processing***, *p.62*

Quantization: Dynamic Levels

The number of dynamic levels is controlled by the ***bit depth***

- Typically 8-, 16-, and 24-bit, corresponding to 2^8, 2^{16}, and 2^{24} levels
- Quantization error between analog value and quantized value

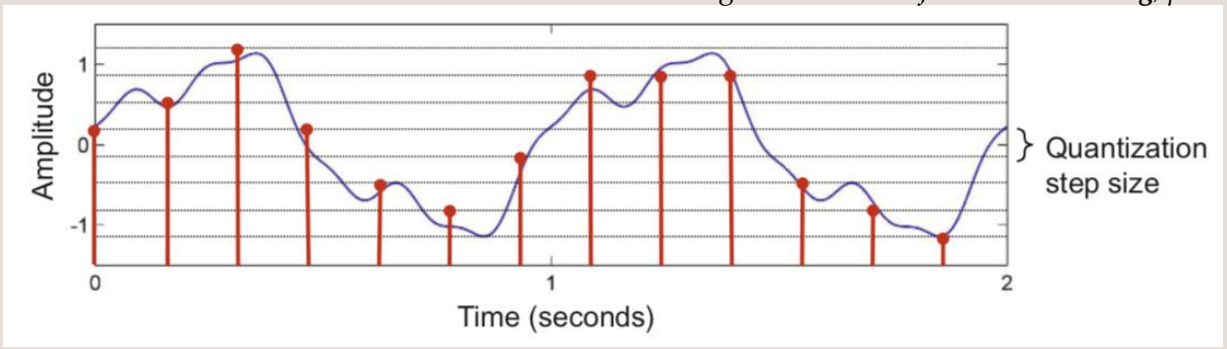
Represented as "***amplitude***"

- Integer vs. float
- Signed vs. unsigned
  - E.g., signed integer 16-bit: -32768 to +32767

Common bit depths

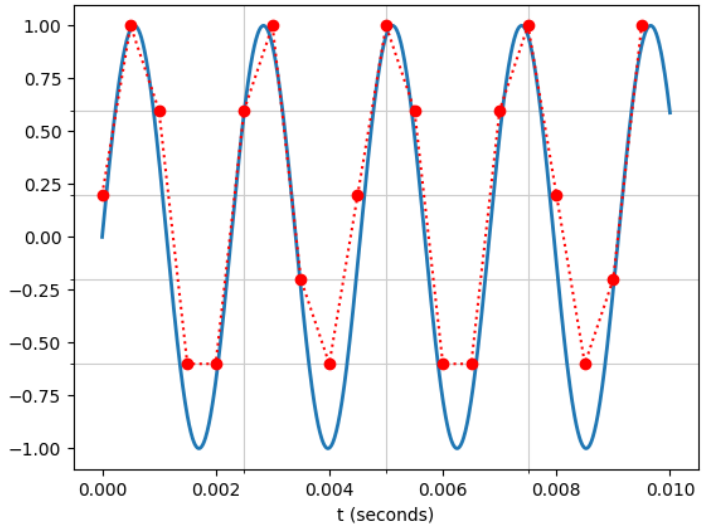
- Audio CD: 16-bit (~96 dB of range)
- Pro audio: 24-bit (~144 dB)

Image from: ***Fund. of Music Processing***, *p.61*

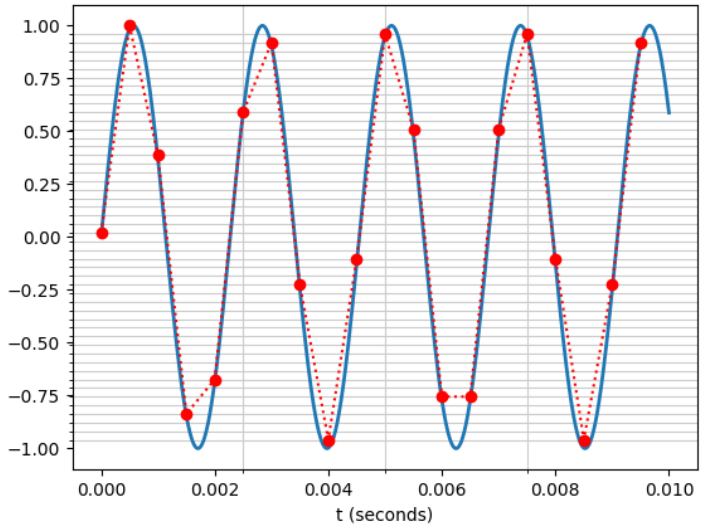


Rounding errors in quantization

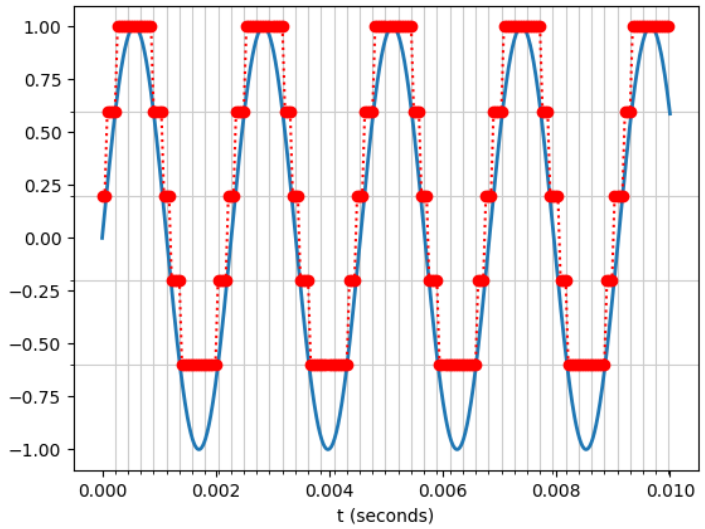
Low bit depth, low sampling rate



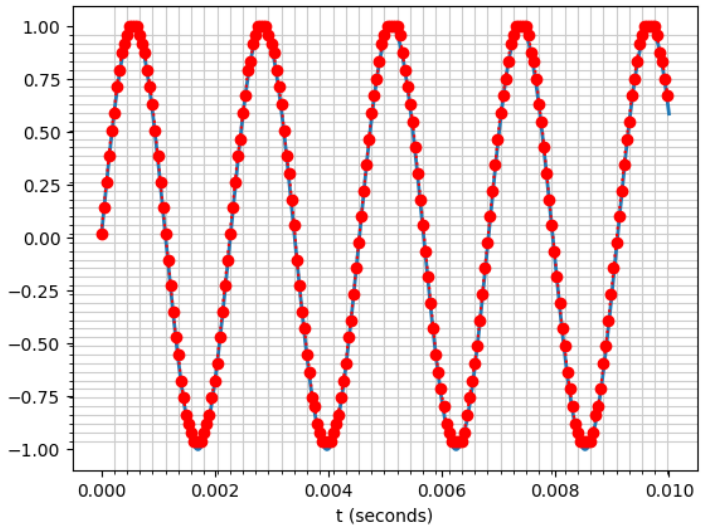
High bit depth, low sampling rate



Low bit depth, high sampling rate



High bit depth, high sampling rate



Audio Channels

Different audio perception on our left and right ear helps determining the ***location*** of the sound source

- "Stereo" sound, with 2 channels
  - Some professional recordings are shaped with *Head Related Transfer Function* (HRTF)



Dummy head microphone for immersive sound experience

Image from: https://lmkprod.com/what-is-hrtf-brief-explanation/

Surround sound systems

- 5.1 or 7.1 with subwoofer (.1)
- 7.1.2 with 2 upward speakers (.2)
- Extra channel → extra storage



HKBU LIATe theatre with 24.2-channel sound system

Image from: http://liate.hkbu.edu.hk

Audio Data Storage

The raw audio data, bit by bit, is called linear pulse-code modulation "linear PCM" encoding (usually as `.wav` files)



- Good enough for sound editing and manipulation
- Consuming too much storage
  - CD quality: 44.1kHz 16-bit 2-channel ➔ 80 minutes consuming ~700 MB

Compressed audio can already fulfil requirements of the general public



- ***Lossy***: MP3, AAC, WMA, …
- ***Lossless***: FLAC, ALAC (.m4a), …
- Audio quality often measured as bandwidth requirement
  - Spotify "normal quality": AAC format at ~96 kbps

Dealing with Music Data: "Symbolic Representation"

Musical Instrument Digital Interface (**MIDI**)

- Standardized in 1983
- Widespread technology to connect various musical hardware and software

In music hardware, MIDI connections can be made with…

- 5-pin MIDI plug
- USB
- Wireless (e.g., Bluetooth)

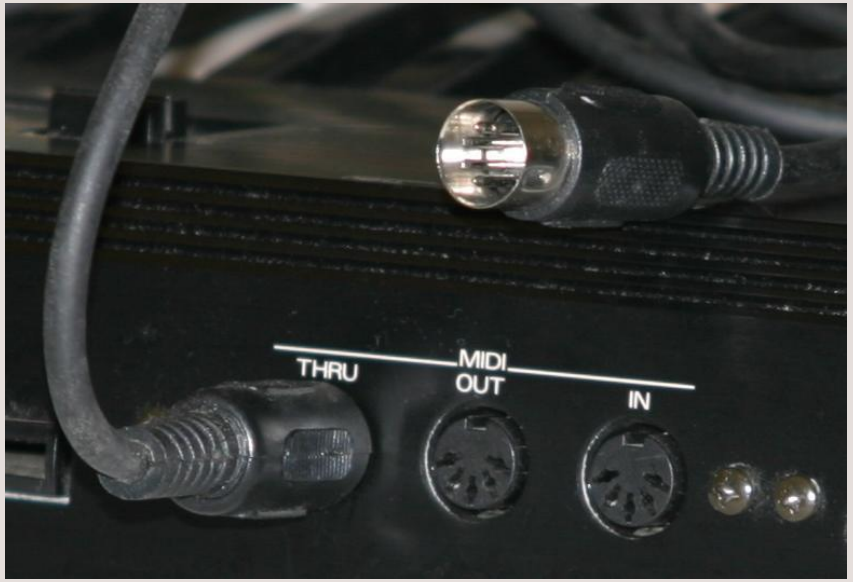


Image from: https://en.wikipedia.org/wiki/MIDI

Dealing with Music Data: "Symbolic Representation"

Control data instead of actual audio data

➔ ***symbolic***

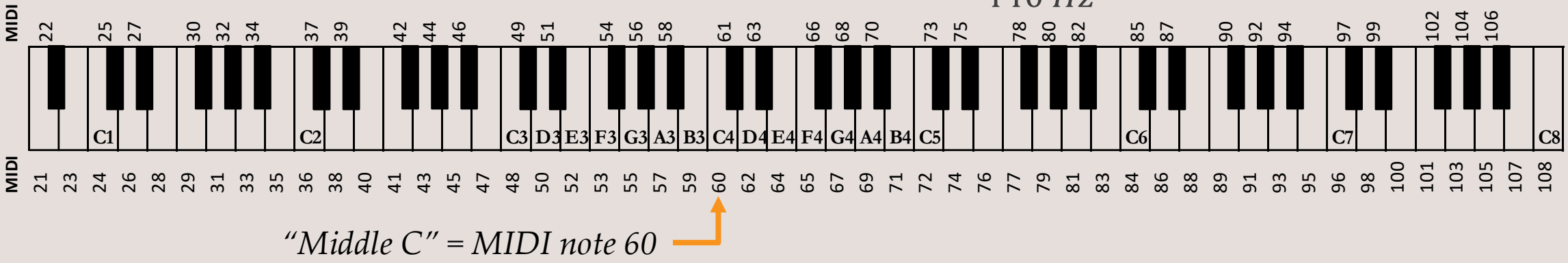
- Playback = *Re-synthesis* on-the-fly
- Basic messages: "**Note on/off, note number, velocity**"
- Only decides when and what should happen, but not HOW
  - Synthesizer will generate the actual sound (e.g., a violin sound, or a piano sound)

MIDI Messages

"**Note On**" denotes the starting point, where "**Note Off**" is to stop

Calculating the ***note number*** (0-127)

$$\text{Note number } p = 69 + 12 \times \log_2\left(\frac{f}{440\ Hz}\right)$$



Velocity (0-127): How fast the key is pressed (correlates to playing intensity)

MIDI Messages

After "*noteon*", the note can be further modified, e.g., with 7-bit values of

- Pitch bend
- Modulation
- Aftertouch/Pressure
- Control change (CC)
  - Damper pedal, sostenuto, expression, portamento, …

As always, the result depends on the synthesizer!

Read more: *https://www.midi.org/midi/specifications-old/category/reference-tables*

MIDI 2.0 (2020): Higher bit precision for fine control of music expression

- *See: https://www.midi.org/specifications/midi-2-0-specifications*

Displaying Music data

Musicians read music, just like we read text

- Different music notation software have their own proprietary standard

An open format: ***MusicXML***

- Storing music notes like MIDI
- Yet it also stores how it should be rendered and displayed
  - E.g., bar lines, key signatures, …



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            <right-margin>752.71</right-margin>
            </system-margins>
          <top-system-distance>170.00</top-system-distance>
          </system-layout>
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        <divisions>1</divisions>
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          <fifths>0</fifths>
          </key>
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          </clef>
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        <voice>1</voice>
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        </note>
      <note>
        <rest/>
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        </note>
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        </barline>
      </measure>
...
```

Musical Instruments

Real World Musical Instruments

How are audio signals generated in real life?

- It's always about vibrations…

Musical instruments are made to create ***regulated*** vibrations

- Usually with a controllable pitch (*periodic waveform*)
- Different instruments show different spectra (*frequency components*)
- The vibration may be amplified with an acoustic body, e.g.
  - Sound box of a violin
  - Sound board of the piano

Real World Musical Instruments

Piano ➔ Chordophones
- Keys driving hammers to hit steel strings and make them vibrate

Strings ➔ Chordophones
- Friction between bow and strings create vibrations

Note: We are not going to cover much physics in this course!

Winds ➔ Aerophones
- Air flow creates vibrations inside the tube of various length

Percussion ➔ Membranophones/Idiophones
- Too many possibilities! In general, an object vibrates when being struck

Electronic music instruments

Ever since the development of electricity, musicians started to explore into new ways of expressing themselves, e.g.,

- Telharmonium (1897)
  - Tonewheels generates electric musical notes with electromagnets
- *Theremin* (1920)
  - Frequency/amplitude of the sound is controlled using the capacitance between hands and antennae
- Analog synthesizers (1960 beyond)
  - Sounds are generated and modified using op-amp, resistors, filters, and other electronics



Video from: https://youtu.be/pSzTPGlNa5U

Augmented Musical Instruments

The mixture of traditional music studies with technologies

- Sensors for more controllability
- Filters for more sound output variety
- Interfaces for expressiveness
- *and a lot more…*

Video from: https://www.youtube.com/watch?v=Ogul65yLBK4



Video from: https://www.youtube.com/watch?v=GB8wtjmOoBY



Music Hardware and Software

Interfacing with Computers

Audio interfaces

- "Sound cards": providing ***signal conversion*** from analog to digital, and vice versa
- DAC + ADC

MIDI interfaces

- Input/output of ***MIDI messages*** between computers and instruments

Control surfaces

- Providing faders and buttons for ***direct control*** over DAW software UI

MIDI controllers

With computers, now we can easily make music with software programs

- MIDI ➔ prominent tool for music production



MIDI controllers are the interface for musicians to make MIDI messages

- Most controllers are in the shape of a piano keyboard
  - 88-key, 72-key, …
  - Weighted keys for realism of touching and expressiveness
- Other shapes
  - Winds, mallet percussion, drums, strings, …



Novel design of instruments

There is a research conference for everything…

- ***NIME*** — The International Conference on New Interfaces for Musical Expression (since 2001): *https://www.nime2024.org*
- Check out publications if you are curious
  - *https://www.nime.org/archives*

Roli Seaboard, LinnStrument, …

- Multi-dimensional, polyphonic interfaces



https://www.nime.org/past_nimes

Music Software

Synthesizers

- Earliest software synthesizers simply emulated the analog synthesizers
- Now there are more complex kinds of synthesis which is only made possible by computer programs
- ***Keyboard workstations*** are controllers with an embedded system for synthesis
- Sometimes exists as plugins for DAWs

Vocaloid

- A special kind of sound synthesizer, which can mimic ***human voice*** by speech synthesis with controls of pitch and other vocal expressions

Music Software

Digital Audio Workstation (DAW)

- Allowing musicians to put together the tasks for ***recording, editing, and producing*** of audio data
- Most of DAW supports VST plugins as a standard way to extend features
- To name a few: *Pro Tools, Logic Pro, Reason, GarageBand*

Live Performance

- Besides using keyboard workstations, a computer with proper software can be used for live performance with maximum flexibility with MIDI controllers and VST plugins
- Examples: *MainStage, Ableton Live*

Music Notation and Recognition

Sheet music is still a major medium among musicians

Music notation software allow easy typesetting with computer keyboard or MIDI controllers

- Examples: *Sibelius*, ~~*Finale*~~, *Dorico*, *MuseScore*
- Some DAWs can also output typeset music

While text can be recognized by OCR technology, handwritten or printed music can also be imported with proper recognition software

- More and more popular on mobile devices

Lecture Review

We have discussed:

- How sounds are produced, and perceived by human
- Ways to digitize sound and music into computer systems
  - Audio vs. symbolic representations
- Music instruments and controllers
- Music hardware and software

Read Further

Section 1.3 "***Audio Representation***", Fundamentals of Music Processing

Chapter 1 "***Audio and Music Signals***", Computer Music Instruments

Section 8.1.1 "***The MIDI protocol***", Computer Music Instruments

- How do computers hear music?
  https://medium.com/@chuckjee/how-do-computers-hear-music-91e699bdf7ea
- More about different kinds of electrified musical instruments
  https://medium.com/@chuckjee/what-is-so-different-when-musical-instruments-are-electrified-a7c62119774c